Why Good Products Go Bad

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The ideas in this article are the author's. A large language model was used to help develop the text and verify references.

Abstract

A pattern repeats across consumer and business technology: a product is excellent when you buy it, and then it degrades through no fault of yours. A home cinema built around component video is stranded by a format transition. A beloved smartwatch dies when its maker is acquired and the servers are switched off. Market-leading virtualization software becomes a billing instrument after its third change of owner. A fitness watch happily exports your data but refuses to import anyone else's. This article names the pattern, catalogs its variants across more than a dozen documented cases, and explains the economic mechanisms behind it: switching costs, lock-in, platform surplus reallocation, planned obsolescence, and ownership changes that hand the customer relationship to owners who bought the lock-in rather than the product. It then examines whether the phenomenon is becoming more common (the defensible answer: the preconditions are measurably rising, even though no study tracks the decay rate itself) and surveys the way out at three levels: what consumers can do before and after purchase, what companies can commit to structurally, and what regulators, particularly in the European Union, are already requiring.

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1 Introduction: Three Purchases and a Pattern

Some years ago I invested in home cinema equipment. The centerpiece was an integrated Onkyo amplifier with support for component video, the three-cable analog standard that carried the best picture quality consumer equipment could produce at the time. I bought a projector matched to DVD resolution and the cables to connect it all. I started collecting DVDs. The system worked, and it worked well. Then Blu-ray arrived. The projector had too little resolution for the new discs, and component video was no longer enough: the industry had moved to HDMI, a digital connection standard with built-in copy protection. The move was not merely a matter of market preference. The licensing rules behind Blu-ray, the compliance rules of AACS (Advanced Access Content System, the copy-protection regime every Blu-ray player manufacturer must license), required players manufactured after 2010 to downgrade their analog outputs to standard definition, and banned analog video outputs entirely on players made after 2013, a schedule known in the industry as the analog sunset [47]. My amplifier did not break. It was made obsolete by contract.

Around the same time I bought a Pebble Steel, the smartwatch from the startup that more or less invented the category. I loved it: a week of battery life, an always-on display, and a thriving app ecosystem. In December 2016 Pebble's founders sold the company's assets to Fitbit for about 23 million dollars [33]. Fitbit kept the servers running for a while, then shut down the app store, voice recognition, and messaging services on June 30, 2018 [54]. Google completed its acquisition of Fitbit in January 2021 [62]. The watch on my wrist was unchanged, but the product I had bought no longer existed, because the product was never just the hardware. It was hardware plus an online service I did not control.

The third case cost businesses far more than it cost me. VMware built the best virtualization software on the market, software that lets one physical computer run many independent virtual machines. It was the market leader on technical merit. Then it passed through a series of owners and, in the judgment of many of its customers, development stagnated. When Broadcom completed its acquisition of VMware in November 2023, in a deal announced at 61 billion dollars in cash and stock plus about 8 billion dollars in assumed debt [8], the squeeze began in earnest: perpetual licenses, which customers buy once and use indefinitely, were discontinued within weeks in favor of subscriptions [4], partner programs were terminated [74], and European customer organizations reported price increases of 800 to 1,500 percent [75]. Nothing about the software got better. Everything about being its customer got worse.

There is also a quieter, everyday variant of the same phenomenon. My Garmin watch synchronizes its health data to Apple Health without complaint. In the other direction, nothing flows: Garmin Connect has for years offered no way to import data from Apple Health or from a competitor's watch [56]. Anyone who has years of training history inside Garmin's service can leave, but their data effectively cannot. Only in 2025 did Garmin announce a partial two-way synchronization, presented in press coverage as letting users switch devices without gaps in their data [7], which is as clear an admission as any of what the one-way design had been costing users all along.

Four stories, one shape: a good product, an investment made in good faith, and a decay triggered not by wear or by my own choices but by decisions made elsewhere, by format cartels, acquirers, and business-model strategists. There is now a word for an important subspecies of this pattern. The writer and activist Cory Doctorow coined *enshittification* to describe the life cycle of online platforms: first the platform is good to its users to build an audience; then it abuses its users to favor its business customers; finally it abuses both to extract value for itself [17]. The American Dialect Society chose it as the 2023 word of the year [3], which says something about how widely the experience is shared. But enshittification, in Doctorow's precise sense, is about platforms. The pattern I keep running into is broader. It includes physical products, open standards replaced by licensed ones, and companies that never ran a platform at all. This article is about that broader pattern.

1.1 What This Article Argues

The argument, in one paragraph: product decay of this kind is not bad luck and not a moral failing of individual companies. It is the predictable result of a specific economic configuration. When customers face high switching costs (the cost of moving to another vendor), the vendor acquires a durable asset: the customers' inability to leave. That asset can be harvested. Harvesting it destroys the product's value slowly and the vendor's reputation eventually, so well-run companies with long horizons often decline to do it. But the asset does not disappear; it waits. An acquisition, a private equity buyout, a new business-model fashion, or a format transition controlled by a licensing consortium can hand the harvesting decision to someone with a shorter horizon and no attachment to the product. Then the harvest begins, and the customer experiences it as a good product going bad.

The article develops this argument in two passes, each organized around the same four questions: why does this matter, what is happening, how does it work, and when and where does it apply, including what follows if nothing changes. The first pass, Part I of the argument (Sections 2–5), covers the phenomenon itself: what forms product decay takes, why the economics produce it, how companies carry it out in practice, and whether it is becoming more common, weighing what the evidence does and does not show. The second pass, Part II (Sections 6–10), covers the remedies: why the market does not correct the problem on its own, what countermeasures exist, how they work at the level of consumers, companies, and regulators, and what happens if they succeed or fail.

Two companion articles on this site approach neighboring questions. *Building Software That Outlives Its Dependencies* [52] treats the producer's side of churn: how to build software when your own vendors and dependencies decay under you. *Capitalism, Markets, and the Missing Objective Function* [53] makes the abstract argument that the market mechanism specifies no social objective, only a procedure, so whatever gets optimized is emergent. This article is, among other things, a concrete instance of that abstraction: a market where the emergent objective has drifted from "serve the customer" to "extract from the customer," one locked-in user at a time.

2 What Is Happening: A Taxonomy of Product Decay

The opening stories are not isolated. Once you have a name for the shape, documented cases turn up everywhere. This section sorts them into five types of decay, distinguished by what triggers the decline. The types overlap in practice (an acquisition often triggers a business-model shift, which is then enforced through a cloud dependency), but separating them clarifies which mechanism is doing the work in each case.

2.1 Standards and Format Churn

The oldest form: your equipment still works, but the ecosystem around it moves on, and sometimes it is pushed. The transition from DVD and component video to Blu-ray and HDMI is the instructive case because it was partly contractual rather than purely technical. The AACS licensing rules that every Blu-ray player manufacturer had to sign required analog video outputs to be degraded to standard definition on players made after December 2010 and removed entirely on players made after December 2013 [47]. Discs could also carry an Image Constraint Token instructing older players to limit their analog output. A consumer who had bought a high-quality analog projector and amplifier did not merely fall behind the state of the art; the state of the art was legally obliged to stop talking to their equipment.

2.2 Acquisition-Driven Decay

The second type begins with a change of ownership. The product is the same on the day after the deal closes; the incentives of its owner are not.

Pebble and VMware, described in Section 1, are both in this class. So is Evernote, the note-taking service, which was acquired by the Italian app company Bending Spoons in early 2023; 129 staff were laid off shortly after the deal, prices rose sharply, and the free tier was limited to 50 notes [30]. So is Skype, though as the passive variant: acquired by Microsoft in 2011 as the world's dominant internet telephony service, it was not so much harvested as neglected for over a decade, then shut down on May 5, 2025 [57]. And so is Revolv, a smart home hub whose maker was bought by Google's Nest division in 2014. In May 2016 Google switched off the servers and every Revolv hub stopped working entirely [39]. Revolv owner Arlo Gilbert's essay about the shutdown became the canonical statement of the problem: "Google is intentionally bricking hardware that I own" [36].

2.3 Business-Model Shifts on Products Already Sold

The third type needs no acquisition. The same company that sold you the product later changes how it wants to make money, and the change reaches backward into what you already bought.

Adobe pioneered the pattern at scale in May 2013 when it ended perpetual licenses for Creative Suite and made future versions subscription-only, over the protest of tens of thousands of customers [2] (existing perpetual licenses kept working; the change closed the road ahead rather than the one behind). Unity, whose game engine thousands of studios had built businesses on, announced a retroactive per-install runtime fee in September 2023; after the developer revolt that followed, the chief executive departed within weeks, and the fee was canceled a year later [86]. BMW sold cars with heated seats installed and then, in 2022, asked owners for 18 dollars a month to switch them on, retreating in 2023 after public ridicule [6]. HP shipped firmware updates (updates to the control software embedded in the device itself) under the name "Dynamic Security" that made its printers reject third-party ink cartridges using non-HP chips, a practice that has produced repeated lawsuits and settlements, including a 2025 settlement in which HP promised to make such firmware updates optional for some models [41]. Philips Hue announced in 2023 that its smart lights, long a flagship example of local, account-free operation, would require a cloud account after all [69]. Chamberlain, maker of the MyQ garage door openers, shut off third-party access through its application programming interface (API), the channel by which other software can control a product, thereby forcing customers who had operated their own doors from home automation software into Chamberlain's own app; the open-source home automation project Home Assistant had to remove its MyQ integration entirely in late 2023 [73].

Sonos deserves special mention because it shows the pattern operating twice on the same customers. In late 2019 the company introduced a trade-in discount whose "Recycle Mode", once activated by the owner to claim the discount, irreversibly bricked (rendered permanently inoperative) the old, still-functional speaker after a 21-day countdown; the scheme was withdrawn in March 2020 after public outcry [83]. In May 2024 Sonos replaced its well-regarded app with a rewrite that removed working features, including alarms and accessibility support. The fallout cost tens of millions of dollars to repair and, in January 2025, cost chief executive Patrick Spence his position [88]. The rewrite was mismanagement rather than deliberate extraction, but it ran on the same tether: the speakers were unchanged, and the product got worse anyway.

2.4 Cloud-Dependency Death

The fourth type is the starkest: hardware you bought outright stops working because a server you never saw is switched off.

Spotify sold a dashboard music device called Car Thing, then announced it would disable every unit in December 2024, initially with no refunds at all [84]; refunds followed after public backlash and the filing of a class-action lawsuit, and the firmware was never released for owners to repurpose the device. Amazon discontinued its Halo fitness trackers in 2023 with the plain statement that, beginning August 1, devices and app would "no longer function" [63]. The smart home company Insteon switched off its servers in April 2022 without any announcement at all; users woke up to dead hubs and a status page still reporting all systems operational [44]. The aftermath, in which the users eventually bought the company itself, is taken up in Section 7.3. VanMoof, the Dutch e-bike maker, went bankrupt in 2023 and left riders with bicycles whose unlocking depended on the company's app and servers [87]. A bicycle is about the last product a customer expects to come with a server dependency, which is precisely why the case is instructive.

Free services follow the same logic without the hardware. Google retired Google Reader in 2013 [1], the moment a generation of users learned that a free cloud service is a revocable privilege. The community-maintained "Killed by Google" catalog lists 307 discontinued Google products as of this writing [50].

2.5 One-Way Data Valves

The fifth type is not an event but a design decision, visible only when you try to leave. A product accepts your data continuously and exports it grudgingly or partially or not at all. Garmin's synchronization with Apple Health, described in Section 1, flows in one direction only [56, 7]. Doctorow described the Kindle in the same terms years earlier, borrowing the slogan of a roach trap: books can check in, but they cannot check out [15]. The same "roach motel" metaphor has been applied to corporate mobile platforms [79]. A one-way valve does not degrade the product today. It quietly raises the cost of leaving, which, as Section 3 shows, is the raw material every other decay type is built from.

2.6 The Catalog at a Glance

Table 1 summarizes the documented cases used in this article, sorted by decay type. Two things stand out from the collection. First, no product category is immune: the table contains amplifiers, watches, speakers, printers, cars, bicycles, garage doors, light bulbs, enterprise software, and game engines. Second, the trigger column shows that the customer's own behavior appears nowhere. In every case the decay was decided somewhere else, which is what distinguishes this phenomenon from ordinary product aging.

3 Why It Happens: The Economics of Harvesting

None of the cases in Section 2 required a villain. Each is the rational output of incentives that economists described decades before the word enshittification existed. This section assembles the mechanism from five well-studied parts.

3.1 Switching Costs Create a Harvestable Asset

A *switching cost* is everything a customer must give up or redo to move from one vendor to another: repurchasing hardware, converting or abandoning data, relearning workflows, rewriting integrations, retraining staff. When switching costs are high, the customer is *locked in*: staying is cheaper than leaving even when the product gets worse or the price goes up. Shapiro and Varian's classic strategy text made explicit what every software vendor understands, that in information markets lock-in is not an accident but something firms deliberately create and value [81]. Farrell and Klemperer's survey of the economics of switching costs states the consequence precisely: firms

Table 1: *Documented cases of product decay, by type. Each case is discussed in the text of this article.*

Product	Decay type	Trigger	Outcome
Component video equipment	Standards churn	AACS analog sunset (2010–2013)	Analog outputs degraded, then banned on new players
Pebble smartwatch	Acquisition	Asset sale to Fitbit (2016)	Cloud services shut down June 2018
VMware	Acquisition	Broadcom acquisition (2023)	Perpetual licenses ended; reported price rises of 800–1,500 %
Evernote	Acquisition	Bending Spoons acquisition (2023)	Layoffs, sharp price rises, free tier cut to 50 notes
Skype	Acquisition	Microsoft acquisition (2011)	Long decline; shut down May 2025
Revolv smart hub	Acquisition, cloud	Nest/Google shutdown (2016)	All units permanently disabled
Adobe Creative Suite	Business model	Subscription pivot (2013)	Perpetual licenses discontinued
Unity engine	Business model	Retroactive runtime fee (2023)	Reversed after developer revolt
BMW heated seats	Business model	Feature subscription (2022)	Reversed after public backlash
HP printers	Business model	“Dynamic Security” firmware	Third-party ink rejected; lawsuits and settlements
Philips Hue	Business model	Mandatory account (2023)	Local-only operation withdrawn
Chamberlain MyQ	Business model	API shutoff (2023)	Home automation integrations broken
Sonos	Business model	App rewrite (2024)	Working features removed; CEO departed
Spotify Car Thing	Cloud dependency	Service termination (2024)	All units bricked; refunds after backlash and lawsuit
Amazon Halo	Cloud dependency	Product exit (2023)	Devices nonfunctional after shutdown
Insteon	Cloud dependency	Unannounced server shutdown (2022)	Hubs dead overnight; later revived by users
VanMoof e-bikes	Cloud dependency	Bankruptcy (2023)	App-dependent unlocking at risk
Garmin Connect	One-way valve	Design decision	Exports to Apple Health; no import path

compete hard for customers ex ante, before the customer is committed, because lock-in gives them lucrative ex post market power afterward, the ability to raise prices or cut quality without losing the customer [32].

This is the load-bearing concept of the whole article, so it is worth restating plainly. A locked-in customer base is an *asset*. It has a market value, it can be bought and sold, and it can be harvested by degrading the ratio of value delivered to price charged. The VMware acquisition is the cleanest illustration: what Broadcom bought for roughly 69 billion dollars was not primarily a codebase; it was tens of thousands of enterprises whose infrastructure could not leave on any short timescale. The reported price increases that followed [75] are simply that asset being converted to cash.

3.2 Platforms Reallocate Surplus Between Sides

Many of the products in Table 1 are *two-sided platforms*: businesses that serve two distinct customer groups who need each other, such as users and app developers, or listeners and advertisers. The economics of such platforms, formalized by Rochet and Tirole, show that the profit-maximizing prices on the two sides are interdependent, and that platforms routinely subsidize one side to attract the other [76]. Nothing in that logic says the subsidy is permanent. Once both sides are aboard and switching costs have accumulated, the platform can reallocate surplus from either side to itself. Doctorow's three-stage enshittification sequence [17] is exactly this reallocation performed in order: subsidize users, then tax the business side, then tax everyone. The sequence itself is not in the theory (Rochet and Tirole's model is a static one), but the theory supplies the pricing freedom the sequence exploits: reallocating surplus to the platform is not a betrayal of the platform model but a move its logic always permitted.

3.3 Planned Obsolescence Is Individually Rational

Planned obsolescence is the deliberate design of products to become useless or unattractive sooner than technology requires. Vance Packard documented the practice, and the advertising culture around it, as early as 1960 [64]. Bulow later gave it a formal economic treatment: a monopolist selling durable goods competes against its own installed base, the products it already sold, and therefore rationally chooses uneconomically short useful lives for them [9]. Strictly, Bulow's result concerns durability chosen at design time by a seller of goods; degrading products already sold goes beyond his model. But the underlying conflict he identified, a vendor competing against its own installed base, is the same, and the software age has made acting on it cheaper than he could have imagined. A firmware update, a dropped API, or a retired server accomplishes overnight what material fatigue used to take years to do, with the added advantage that the manufacturer can time it precisely. The analog sunset of Section 2.1 shows the same logic operating at the level of an industry consortium rather than a single firm: the useful life of every component video installation was ended by license terms, on a schedule, industry-wide.

3.4 Acquisitions Transfer the Harvest Decision

If lock-in is an asset, why do healthy companies often leave it unharvested for years? Because harvesting is irreversible: it burns reputation and product quality, which are worth more to an owner with a long horizon than the one-time extraction is. The founders of a company usually carry attachments that make harvesting unattractive: pride in the product, relationships with early customers, a planned future in the market.

An acquisition severs those attachments in a single transaction. The buyer did not build the product and does not answer to its history; it paid for the discounted value of the customer base and needs to recover the price. This is why the ownership-change cases in Table 1 cluster so tightly. The private equity industry has refined the approach into a repeatable playbook of debt loading, fee extraction, and quality cuts, documented at book length by Ballou [5], but the mechanism

does not require private equity. Any buyer whose spreadsheet values the installed base above the product's future will behave the same way. Christensen's account of the innovator's dilemma adds the passive variant: incumbent market leaders can stagnate honestly, by rationally serving their existing customers while the technology moves on, no hostile owner required [10]. VMware arguably suffered both in sequence: years of incumbent stagnation under earlier owners, then active harvesting under Broadcom.

3.5 The Constraints That Used to Prevent This

The parts assembled so far explain why harvesting is profitable. They do not explain why it has become so visible now, since switching costs and acquisitions are hardly new. Doctorow's answer is that four external constraints used to make harvesting dangerous, and all four have eroded: competition (antitrust enforcement weakened for decades, so customers have fewer places to flee), regulation (platforms outgrew the rules written for earlier industries), interoperability (laws against circumventing digital locks made it illegal to build the adapters and compatible products that once disciplined vendors), and worker power (employees who once refused to degrade their own products lost the leverage to refuse) [19, 20]. Section 5 examines how much measurable support this erosion story has, at least for the constraints that can be measured.

The companion article on capitalism's missing objective function [53] supplies the frame that ties the five parts together. The market mechanism specifies procedures, prices, property, and voluntary exchange, but no social objective; what gets optimized is emergent. For products under lock-in, the emergent objective drifts in a predictable direction: the sunk investments of existing customers appear in no company's objective function, except as an asset to be harvested. Nothing in the mechanism pushes back. That is not a market failure in the textbook sense. It is the market working exactly as specified, on a specification that never mentioned the customer's amplifier.

4 How They Do It: The Playbook

The economics of Section 3 are carried out through a small repertoire of concrete techniques. Learning to recognize them matters for a practical reason: most of them are visible, or at least inferable, before you buy. This section names the seven that recur across the case catalog.

Firmware as a weapon. A software update changes what the hardware you own will do, after you own it. HP's "Dynamic Security" updates taught printers to reject working third-party cartridges [41]. The same channel that delivers security fixes delivers new restrictions, which is why update policy is a trust question, not merely a maintenance question.

The forced account. A product that worked locally acquires a mandatory cloud login, converting an anonymous owner into a managed, monetizable user, as Philips Hue did in 2023 [69]. The account is the hinge on which later degradation swings: terms can change per user, features can be moved behind tiers, and the product can be disabled remotely.

The API shutoff. The API through which other software controls a product is closed to force all use through the vendor's own app, as Chamberlain did with MyQ [73]. Interoperability is removed retroactively from devices customers already installed.

The license switch. Perpetual licenses become subscriptions, converting a one-time purchase into a permanent revenue stream and a permanent dependency: stop paying and the product stops working. Adobe executed the switch for creative software in 2013 [2]; Broadcom executed it for VMware within weeks of closing [4]; Unity attempted a retroactive variant and failed [86].

The server shutdown. For any product with a cloud dependency, turning off the servers is the cheapest possible end-of-life plan, and the cost lands entirely on the customer: Revolv [39], Car Thing [84], Halo [63], Insteon [44]. The technique requires no engineering at all, only the absence of an obligation to do anything else.

Parts pairing. Software serialization binds individual components to an individual device so that a functionally identical replacement part is rejected unless the manufacturer blesses it. The technique converts repair into a licensed activity. It was significant enough that Oregon's 2024 right-to-repair law specifically banned it [72].

The one-way valve. Data import paths are omitted while export paths to bigger ecosystems are maintained, as in Garmin Connect [56]. Unlike the other six techniques, this one is passive: it does nothing to today's product and everything to tomorrow's exit.

The list rewards a moment of reflection at purchase time. For any product you are considering, ask which of the seven techniques its vendor *could* apply later: Does it update its own firmware? Does it work without an account? Is the API contractual or tolerated? Is the license perpetual? Does it function without the vendor's servers? Can parts be replaced generically? Does data come out as easily as it goes in? The questions have factual answers, and Section 7 turns them into a buying checklist.

5 When and Where: Is It Getting Worse?

Anyone who has lived through the cases in this article's catalog will feel that the phenomenon is accelerating. Feelings are not evidence, and this section separates what is measured from what is merely salient.

5.1 What the Evidence Shows

Start with the honest negative: no study I could find measures the thing directly. There is no time series of "products degraded or bricked per year," and the growing catalog of incidents is contaminated by a visibility bias: cloud-connected products barely existed before roughly 2010, so "more cloud-dependent products are dying" is partly just "more products are cloud-dependent." The claim that decay events themselves are becoming more frequent cannot currently be backed by data, and this article does not make it.

What can be backed by data is that every precondition of the harvesting mechanism in Section 3 is rising.

Market power. Average markups across United States firms, the gap between prices and marginal cost, rose from about 21 percent above cost in 1980 to about 61 percent in 2016 [13]. Degrading a product without losing its customers is far safer where competition is weak, and markups are a widely used, though contested, indicator of weakening competition (part of the measured rise may reflect growing fixed costs rather than market power). The same reversal shows at the level of whole markets: American industries have grown more concentrated and entry has fallen since the 1990s [68].

Fewer escape routes. The share of United States firms that are newly founded fell from roughly 12 percent in the late 1980s to below 8 percent in the wake of the Great Recession, part of a broader decline in business dynamism, the rate at which firms enter, grow, and exit markets [14]. There is also a documented "kill zone" effect: after the largest platforms buy a startup (the study covers Google and Facebook acquisitions), venture funding in that startup's space falls sharply [48]. Both matter here because harvesting a locked-in base is only safe when no fresh alternative is waiting to receive the refugees.

The legal shift from owning to licensing. License agreements have systematically displaced personal property rights in digital goods: the thing you "bought" is, in the fine print, a revocable permission [67]. This is the legal substrate that makes the server shutdown and the license switch lawful.

The subscription and cloud shift. Gartner projected that more than half of enterprise information technology spending in key market segments, application software chief among them, would flow to the cloud by 2025 [35]. Zuora's Subscription Economy Index, which should be read with the caveat that Zuora sells subscription billing software and measures a self-selected cohort, reports subscription businesses growing several times faster than the S&P 500 over the last decade [89]. Every percentage point of that shift converts owned products into tethered ones.

Opacity at the point of sale. In 2024 the United States Federal Trade Commission (FTC) surveyed 184 connected consumer products and found that about 89 percent of their manufacturers' web pages did not disclose how long software support would last [82]. The one fact a buyer would need in order to price the risk this article describes is, in nine cases out of ten, not published.

Mapped back onto Doctorow's four constraints, this evidence covers competition well, the regulatory environment through the licensing shift, and interoperability only partly, eroded as it is by the same contractual turn. The fourth constraint, worker power inside product companies, has no measured counterpart that I could find, and that should be said plainly rather than counted as confirmed.

5.2 Where It Strikes

The mechanism predicts its own geography. Decay concentrates where switching costs are high and alternatives are few: ecosystems with accumulated data (fitness history, photo libraries, note archives), infrastructure with high migration costs (virtualization, enterprise software), devices tethered to vendor servers, and markets consolidated down to two or three players. It is rare where products are commodities: nobody enshittifies a hammer, because its owner can replace it in an afternoon with any other hammer. The practical corollary: the more a product asks you to invest in it, the more skeptically its exit paths deserve to be examined.

5.3 What If Nothing Changes

Project the preconditions forward and the endpoint is easy to describe, because parts of it have already arrived. Durable goods stop being durable in any meaningful sense: their lifetime becomes $\min(\text{hardware life}, \text{vendor patience})$, and the FTC survey shows vendor patience is not even disclosed. Ownership becomes a euphemism for the front-loaded payment plan of a service the seller can repossess by inaction. Perfectly functional hardware flows into electronic waste on the vendor's schedule rather than the laws of physics. And a subtler cost compounds underneath: rational customers learn to withhold trust, to avoid ecosystems, integrations, and up-front investments, because experience has taught them that today's excellent product is collateral for tomorrow's harvest. Markets run on the expectation that quality is rewarded; a market where the rational strategy is defensive buying has broken that link. An optimization process with no term for customer welfare will, given time and lock-in, keep discounting it toward zero. The second half of this article is about the forces pushing back.

6 The Way Out: Why Remedies Are Needed, and What They Are

The natural objection to everything so far is that markets are supposed to punish this behavior on their own. Customers who get burned leave; reputations follow companies around; competitors advertise against the incumbent's misdeeds. Why does the correction not simply happen?

6.1 Why the Market Does Not Self-Correct

Three reasons, each already established in Part I.

First, exit is muted by construction. The whole point of switching costs is that leaving hurts more than staying, even as quality falls (Section 3.1). Punishing VMware by migrating a thousand virtual machines is a multi-year project; punishing Garmin means abandoning my decade of training data. The customers most able to punish a vendor are precisely the ones the lock-in was designed to hold.

Second, the information needed to punish in advance is missing. A market disciplines bad behavior partly through buyers who avoid it beforehand, but a buyer cannot price a risk that is not disclosed: nine out of ten connected products do not state how long they will be supported [82], and no product ships with a label saying "the servers this depends on are a cost center under quarterly review." The decay usually arrives years after the purchase, long after the price signal could have done its work.

Third, each individual harm is too small to litigate and too diffuse to organize. A bricked 90-dollar gadget is not worth a lawsuit; a thousand of them are a class action at best, and the Car Thing case shows that even then the remedy arrives late and partial [84].

Hirschman's classic framework gives the vocabulary for what remains. Customers of a declining organization have three options: *exit* (leave for an alternative), *voice* (protest and push for change from within), and *loyalty* (stay and hope), and the interesting dynamics come from how the first two interact [40]. Lock-in is, in Hirschman's framework, a tax on exit. Where exit is taxed, voice is the channel that remains, though Hirschman himself warned that voice borrows much of its power from the credible threat of exit, a warning Section 7 confirms in practice. Where both fail, the remaining moves are structural: change what companies are, or change what the law requires. That observation organizes everything that follows.

6.2 The Remedy Map

The remedies that exist operate at three levels, and the next three sections take them in turn.

Consumers (Section 7) can restore their own exit option before purchase, by selecting products whose architecture makes harvesting difficult, and can exercise voice after it, which the record shows works more often than cynicism predicts.

Companies (Section 8) can make commitments that outlast their own future management: open protocols, longevity as a business model, ownership structures that cannot be bought, and a graceful death for products they abandon.

Regulators (Section 9) can force the disclosures and exit paths the market fails to produce, and in the European Union have already begun to, with obligations that reach from repairability through update lifetimes to data portability.

None of the three levels suffices alone. Consumer discipline fails without the information regulation forces into the open; regulation lags years behind practice; company-level commitments are only credible when customers reward them, which requires consumers who know to look. The levels are complements, not alternatives.

7 How Consumers Can Defend Themselves

The consumer's leverage is concentrated at two moments: before the purchase, when every option is still open, and during a public backlash, when the vendor's calculation can still be changed. This section covers both, plus the last resort.

7.1 Before You Buy: The Checklist

Section 4 listed the seven techniques of decay. Most have a purchase-time counter, and together the counters form a checklist.

- **Prefer local control.** A device that works without the vendor's servers cannot be killed by them. This property is now easy to select for in at least one market: the Matter smart home standard, launched in 2022 by an industry alliance, is designed so that device control runs over the local network rather than through a vendor's cloud [11].
- **Prefer open standards over licensed connectors.** The analog sunset was possible because every player manufacturer had signed the same license. A product speaking open protocols has no license to be sunset by.
- **Check the data exits before you enter.** Before committing years of data to a service, verify that a complete, documented export exists, and distrust one-way valves on principle (Section 2.5). Bulk export tools such as Google Takeout, which by Google's own account handles millions of exports per month [38], and the cross-vendor transfer tools of the Data Transfer Initiative, a nonprofit founded by Apple, Google, and Meta [12], exist because portability is technically easy; a vendor that does not offer it has chosen not to.
- **Avoid forced accounts where a product does not need them.** An account is remote-degradation infrastructure (Section 4).
- **Check repairability before price.** iFixit publishes teardown-based repairability scores that measure exactly the properties (generic parts, published manuals, absence of software locks) that determine whether hardware outlives its vendor [42].
- **Judge the update channel by its history.** Firmware updates and API access can be revoked after purchase (Section 4); a vendor that has already used updates to remove capability, or closed an API on its installed base, has told you what its channel is for. Prefer vendors whose interfaces are documented and contractual rather than merely tolerated.
- **Prefer purchases that survive the store.** For media and games, offline installers without digital rights management (DRM), the copy-protection software that ties a purchase to a vendor's authorization servers, mean the purchase outlives the storefront. GOG, an online storefront for DRM-free games, makes this an explicit product promise through its preservation program [37].

For software, the strongest architectural version of the checklist has a name: *local-first software*, defined by Kleppmann and coauthors as software where the primary copy of your data lives on your own device, with the cloud reduced to optional synchronization, so that the product keeps working when the vendor's servers do not [51]. And the fully sovereign end of the spectrum, self-hosting, running the service side yourself on your own hardware, has quietly left the hobbyist niche: the Home Assistant project reports over 2 million active installations of its self-hosted home automation platform [78].

7.2 After You Buy: Voice Works More Often Than Expected

The record in Table 1 contains more reversals than a cynical reading predicts. BMW dropped the heated-seat subscription [6]. Unity canceled the runtime fee and lost its chief executive on the way [86]. Sonos scrapped Recycle Mode [83] and later paid for the app disaster with its leadership [88]. Spotify paid refunds once a class action made bricking expensive [84]. The pattern behind the reversals: voice succeeds when it is loud enough to threaten the vendor's reputation among customers not yet locked in, whom the vendor still has to court. This is

Hirschman's warning from Section 6 confirmed: voice worked in each case because somebody's exit was still credible.

Organized voice scales further. The Stop Killing Games campaign, organized as a European Citizens' Initiative (a formal EU petition mechanism that obliges the European Commission to respond once one million verified signatures are collected), was launched after Ubisoft shut down the servers of a paid game. It gathered 1.4 million signatures in 2025 for the demand that games sold to consumers must be left in a functional state, by patch or by released server software, when support ends [65]. Whatever its regulatory fate, it demonstrated that end-of-life obligations, the core idea of Section 9, can command mass public support.

7.3 The Last Resort: Community Rescue

When exit and voice both fail, communities have repeatedly performed a third maneuver Hirschman did not name: taking over the product. After Fitbit shut down Pebble's services, the volunteer Rebble project rebuilt the server side and has kept the watches functional since [71]. When Insteon's servers went dark, a group of users eventually bought the company itself and turned the servers back on [43]. These rescues are inspiring and should not be romanticized: they depend on an unusually capable community, on reverse engineering being feasible, legal, or tolerated, and on luck. They are the exception that proves why the structural remedies of the next two sections matter, since a well-designed product would not need rescuing.

8 How Companies Can Avoid It

A company that intends to treat its customers well faces a credibility problem: intentions do not survive acquisitions, and Section 3.4 showed that the decision to harvest is typically made by a future owner the current management will never meet. The remedies that work at the company level are therefore *structural*: commitments embedded in architecture, contracts, or ownership, where a future owner cannot simply reverse them.

8.1 Architecture as Commitment

The cheapest structural commitment is technical. A product built on open protocols with local control, in the sense of Section 7.1, is one whose customers a future owner cannot hold hostage, because the exit door is welded open. Matter certification is one legible form of this promise [11]. Documented data export is another. From the vendor's side these look like concessions; they are better understood as a bond posted against the vendor's own future misbehavior, and customers who have read Section 4 increasingly price that bond in.

A stronger variant makes longevity itself the business model. Fairphone sells phones with a five-year warranty and software support guaranteed into the next decade (eight years, with an aim of ten) [31]. Framework sells laptops designed around user-replaceable modules, with every spare part sold openly and repair guides published [34]. Both companies profit from exactly the property (long product life) that Bulow's monopolist rationally destroys [9], which is possible because their customers select them for it. The business model is the commitment.

8.2 Ownership as Commitment

Architecture cannot prevent an acquirer from discontinuing the product outright. For that, the ownership itself has to be exit-proof.

The plainest form is the foundation. The Signal messenger is operated by a nonprofit foundation, launched in 2018 with 50 million dollars from a WhatsApp co-founder, precisely so that the service cannot be acquired and monetized against its users [55]. The Home Assistant project went

further in 2024, transferring itself and over two hundred related projects into the Swiss Open Home Foundation, created, in the foundation's own words, to protect them against "surveillance capitalism, the risk of buyout, and open source projects becoming abandonware" [61]. A more general corporate form, *steward-ownership*, separates voting control from profit rights so that the company can never be sold as a financial asset at all [70]. What these structures share is that they remove the transaction of Section 3.4 from the space of possible futures: there is no one to sell the locked-in customers to.

8.3 Death as Commitment

Every product eventually reaches end of life; the structural question is who pays for the landing. Two practices convert a shutdown from a harvest into a handover.

The first is *source-code escrow*, a contractual arrangement, long standard in business-to-business software, in which the vendor deposits its source code with a neutral third party that releases it to customers if the vendor fails or abandons the product; the practice is now being extended to keep cloud services running after vendor failure [49]. Consumers are almost never offered an equivalent, and there is no technical reason for that gap.

The second is the *graceful death*: releasing the code at shutdown so the community can carry the product. The precedents exist. When Lowe's closed its Iris smart home platform in 2019, it refunded hardware and open-sourced the platform software for self-hosting [45]. And in January 2025, Google, of all the actors in this article, open-sourced PebbleOS, the operating system of the watch its own acquisition chain had killed [46], enabling Pebble's founder to launch new hardware on the revived code. The acquirer that ended the product later enabled its resurrection, which demonstrates both that the graceful death costs the vendor almost nothing and that it can arrive seven years too late to be graceful.

Doctorow's structural proposal generalizes all of this: *interoperability*, the ability of other parties' products to connect to yours and work with it, ideally protected in law, keeps every vendor honest by keeping every customer mobile. Where vendors refuse, *adversarial interoperability*, building the compatible product without permission, has historically done the same job [16, 18]. A company that embraces interoperability voluntarily is making the one commitment that subsumes the others: it is agreeing to be judged by its product rather than by its exit tax.

9 How Regulators Are Forcing the Issue

Regulation addresses the two failures the market cannot fix from inside: missing information at purchase time and missing exit paths afterward. The European Union has moved furthest, and the dates matter, because several of the obligations are arriving right now. This article is dated July 31, 2026, which happens to be the deadline by which every EU member state must apply the Right to Repair Directive. The regulatory landscape described below is not a proposal; most of it is law.

9.1 Repair and Product Lifetime

Directive (EU) 2024/1799, the Right to Repair Directive, obliges manufacturers of products in covered categories, including phones, tablets, and major appliances, to repair them at a reasonable price and within a reasonable time, even outside the warranty period; member states must apply it from July 31, 2026 [27]. A *directive*, in EU law, binds member states to transpose rules into national legislation, as opposed to a *regulation*, which applies directly.

Product-design rules reach further back in the pipeline. Since June 20, 2025, smartphones and tablets sold in the EU must receive operating system updates for at least five years after the last unit of a model is sold, and critical spare parts must remain available for seven years after

discontinuation, under Commission Regulation (EU) 2023/1670 [21, 22]. The newer Ecodesign for Sustainable Products Regulation provides the framework for extending such requirements (durability, repairability, and software support among them) to further product categories [28]. France, ahead of the curve, has required a repairability score on electronics price tags since 2021 and began replacing it with a broader durability index in 2025 [60, 80].

Two more instruments close specific gaps from Part I. The Sale of Goods Directive requires sellers of goods with digital elements to supply the updates, including security updates, necessary to keep the product functioning as sold [24], which makes the degrading firmware update of Section 4 a conformity problem rather than a vendor prerogative. And the Cyber Resilience Act requires manufacturers of products with digital elements to provide security updates for the product's expected lifetime, and for at least five years unless the product is expected to be used for a shorter time, and, critically, to *disclose the support period*, with full application from December 2027 [29]. That disclosure is aimed squarely at the information failure the FTC survey documented [82]: the support lifetime becomes a label a buyer can compare, and therefore a property vendors can compete on.

In the United States, progress runs state by state. New York's Digital Fair Repair Act of 2022 was the first state law requiring manufacturers to give independent repairers the same parts, tools, and documentation as their authorized channels [59]; Oregon's 2024 law was the first to ban parts pairing outright [72]. Perzanowski's book-length treatment makes the general case that repair is a property right worth restoring, not a market niche [66].

9.2 Data Portability: Dismantling the One-Way Valve

Three EU instruments attack the one-way valve of Section 2.5, with increasing force.

The General Data Protection Regulation's Article 20 grants individuals the right to receive their personal data in a machine-readable format and transmit it elsewhere [23]. It is a snapshot right, exercised request by request rather than as a live feed, and it covers data the user provided (including observed usage data) rather than what was derived from it, but it establishes the principle.

The Digital Markets Act tightens the obligation for *gatekeepers*, the handful of the largest platform companies that the European Commission formally designates: Article 6(9) requires them to provide continuous, real-time data portability, free of charge [25]. Garmin is no gatekeeper, so this instrument does not reach the case that opened this article.

The Data Act does. Regulation (EU) 2023/2854, applicable since September 12, 2025, grants users of *connected products*, hardware that generates data about its use, the right to access the readily available data their product generates and to share it with third parties of their choosing (with safeguards for trade secrets, and with designated gatekeepers excluded as recipients) [26]. A fitness watch is a connected product. The regulatory intent is the inversion of the roach motel: data that checks in must be able to check out, to a destination the user picks. How much practical portability this yields will depend on enforcement, but the legal default for the exit side has flipped: refusing the user access to their own data is no longer a design choice a vendor in the EU market simply gets to make. What no law yet requires is the mirror image, an obligation to *accept* imported data, and that is the half of the valve the Garmin case turns on. Regulation is prying open the exit; the entrance remains at the vendor's discretion.

9.3 The Missing Piece: End-of-Life Obligations

What no jurisdiction yet requires is the graceful death of Section 8.3: an obligation, when a server-dependent product is discontinued, to patch it for standalone operation or release the server components. The Stop Killing Games initiative put precisely that demand in front of the European Commission with 1.4 million signatures behind it [65]. Whether or not that particular

campaign produces legislation, it marks where the regulatory frontier currently runs: purchase-time disclosure and repair are becoming law; the right to a product that outlives its vendor is still being argued.

10 What If the Remedies Take Hold

The remedy map of Sections 7–9 is not hypothetical; enough of it is running that we can observe what success and failure look like.

10.1 Signs That the Direction Works

The rescues already described, Rebble, the Insteon buyout, and the rePebble hardware built on freed code (Sections 7.3 and 8.3), are the existence proof, and they establish a counterfactual for the whole catalog: the value destroyed in a shutdown is not gone because the product stopped being viable; it is gone because the shutdown was the cheapest option available to the vendor. When someone else is allowed to pick the product up, it turns out to still work.

At institutional scale, the German state of Schleswig-Holstein is migrating 30,000 government computers from the Microsoft stack to Linux and LibreOffice, explicitly to escape vendor dependence [85, 77]. Whatever one thinks of the specific trade-offs, it demonstrates that even the heaviest lock-in, a government's document infrastructure, is escapable when an organization decides exit is worth funding. Exit, once demonstrated, disciplines vendors far beyond the customers who actually leave; the credible threat is the mechanism.

If the regulatory arc of Section 9 holds, the realistic good outcome is not a world without product decay. Companies will still fail, formats will still turn over, and acquirers will still harvest what they can. The realistic good outcome is a market where the decay risk is *visible and priced*: support lifetimes on the label, repairability on the price tag, data required to be portable, and a growing segment of vendors competing on longevity because a checklist like Section 7.1 has enough users to be worth winning. Decay would stop being a trap and become a disclosed product attribute, like fuel consumption.

10.2 The Limits

The failure modes deserve equal time.

Regulation is slow and territorial. The EU rules bind the EU market; customers elsewhere benefit only where global product lines make compliance cheaper than segmentation. Enforcement lags text: the Data Act's portability rights are law, but their practical force against a determined vendor is untested.

Voice is fickle. For every reversed decision in Section 7.2 there are quiet degradations that drew no headlines and stood. The Stop Killing Games initiative gathered its 1.4 million signatures [65], and still faces the gap between a demonstrated public demand and an actual legislative act.

Community rescue does not scale. It requires an unusually devoted user base, tractable technology, and a tolerant legal environment; VanMoof riders and Halo owners got no Rebble. Even consumer-advocacy infrastructure itself is subject to the phenomenon this article describes: Mozilla's *Privacy Not Included* buyer's guide, one of the few systematic attempts to evaluate products on exactly these risks, has been effectively dormant since Mozilla dissolved the advocacy division that produced it in late 2024 [58].

And the deepest limit: every remedy in this article raises the cost or lowers the yield of harvesting, but none abolishes the underlying asset. As long as switching costs exist, locked-in customers remain a thing that can be bought, and Section 3.4 will keep finding buyers. The remedies do not end the game. They change the odds, which, for consumers who have spent two decades on the losing side of it, would already be a great deal.

11 Conclusion

A product, it turns out, is a bundle of promises: that the hardware will keep doing what it did in the demo, that the servers behind it will stay up, that the data going in can come back out, that the license you paid for means what it appeared to mean. The market prices the object precisely and the promises hardly at all, because the promises break years after the money changes hands, usually under a different owner. This article has argued that the breaking of those promises, the thing that stranded my amplifier, turned my watch into a paperweight, and made VMware a cautionary tale, is one phenomenon with one economic engine, wearing different costumes.

The first half of the argument: switching costs turn a customer base into a harvestable asset (Section 3); the harvest is carried out through a small, recognizable playbook of firmware updates, forced accounts, license switches, and server shutdowns (Section 4); and while nobody measures the decay rate directly, every precondition of it (market concentration, subscription tethering, cloud dependency, undisclosed support lifetimes) is measurably rising (Section 5). The pattern is not misconduct at the margin of the market. It is the market optimizing exactly what its mechanism rewards, with no term in the objective for the customer already inside.

The second half: the correction does not come from the market alone, because lock-in mutes exit and the necessary information is hidden at purchase time (Section 6). It comes, where it comes at all, from three directions at once. Consumers can select for architectures that keep the exit open and can raise the cost of decay when it arrives (Section 7). Companies can bind their future owners with open protocols, longevity business models, foundation ownership, and a commitment to die gracefully (Section 8). Regulators, above all in the European Union, are beginning to force the disclosures and the portability the market never produced on its own (Section 9).

For the reader who wants the operational residue of this article in six lines, here is the buying checklist of Section 7.1, compressed and with one closing question added. It is the list I now apply, and wish I had applied before the Onkyo:

- Does it work, fully, without the vendor's servers and without an account?
- Does it speak open standards, or a licensed connector that a consortium can sunset?
- Can every important piece of data be exported completely, today, in a documented format?
- Is the license perpetual, and if not, what exactly stops working when the payments do?
- Can it be repaired with generic parts, and does anyone publish the score?
- If the vendor were acquired tomorrow by its least sentimental possible buyer, what would that buyer do with what you are about to hand over?

The last question is the whole article in miniature. You are never only buying a product. You are choosing a counterparty for a relationship whose terms the other side can rewrite, and the discipline that keeps those terms honest (competition, interoperability, regulation, and informed exit) is exactly what two decades of consolidation have worn down and what the remedies of Part II are slowly rebuilding. Buy, build, and legislate accordingly.

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Change Log

- **July 31, 2026**: First version.
- **August 1, 2026**: Added a disclosure note about language-model assistance.